

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

ANTI-LOCK BRAKE SYSTEM CONTROL MODULE (ABS) - HYBRID ELECTRIC VEHICLE

WARNINGS:

- A risk assessment must be performed before any work is undertaken.
- All hybrid electric vehicle work must be performed by a suitably qualified person.
- Appropriate Personal Protection Equipment (PPE) must be worn when working on or near a hybrid electric vehicle high voltage system.
- The hybrid electric vehicle battery pack casing must not be opened due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- The electric power inverter converter casing must not be disassembled due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- If the hybrid electric vehicle battery pack is damaged or overcharged, there is a risk of exposure to hazardous voltage and/or highly corrosive electrolyte mist. If liquid or vapour is observed leaking from the hybrid electric vehicle battery pack,

take the following action:

- Evacuate the area
- Notify manager
- Do not breathe smoke/vapour
- Contain spillage using spill kit
- Wash any spillage off body and clothing (remove contaminated clothing)
- If the hybrid electric vehicle warning indicator is illuminated, the Battery Energy Control Module (BECM) cannot isolate the high voltage cables. Live working will be necessary to rectify the fault.
- The motor generator can generate a voltage if the wheels are rotated, even if the hybrid electric vehicle system has been made safe.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Anti-lock Brake System Control Module (ABS) - Hybrid Electric Vehicle. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Braking Control System - SDV6 3.0L](#)

Diesel - Hybrid Electric Vehicle (206-11 Brake Controls, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0030-38	Left Front Tone Wheel - Signal frequency incorrect	- Front left wheel speed sensor reluctor ring damaged - Front left wheel bearing failure - Front left wheel/tire size incorrect	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Check the integrity of the front left wheel speed sensor reluctor ring - Check the front left wheel bearing - Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the front left wheel and tire for condition and balance. Check and tighten the wheel nuts
C0031-11	Left Front Wheel Speed Sensor - Circuit short to ground	Front left wheel speed sensor power circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground
C0031-12	Left Front Wheel Speed Sensor - Circuit short to battery	Front left wheel speed sensor power circuit short circuit to power	Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to power
C0031-14	Left Front Wheel Speed Sensor - Circuit short to ground or open	Front left wheel speed sensor signal circuit short circuit to ground, open circuit, high	Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, open circuit, high resistance. Using the

		resistance	manufacturer approved diagnostic system, check datalogger signal - Left Front Wheel Speed Sensor Input (0x2B06)
C0031-1C	Left Front Wheel Speed Sensor - Circuit voltage out of range	Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0031-25	Left Front Wheel Speed Sensor - Signal shape/waveform failure	- Front left wheel speed sensor signal circuit short circuit to ground - Front left wheel speed sensor failure	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor
C0031-29	Left Front Wheel Speed Sensor - Signal invalid	- Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front left wheel speed sensor signal circuit short circuit to	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit

		ground, short circuit to power, open circuit, high resistance	diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0031-2F	Left Front Wheel Speed Sensor - Signal erratic	Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0031-31	Left Front Wheel Speed Sensor - No signal	Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance

C0031-4A	Left Front Wheel Speed Sensor - Incorrect component installed	Incorrect front left wheel speed sensor installed	Install the correct front left wheel speed sensor
C0031-62	Left Front Wheel Speed Sensor - Signal compare failure	- Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0031-66	Left Front Wheel Speed Sensor - Signal has too many transitions/events	- Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front left wheel/tire size incorrect	- Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the front left wheel and tire for condition and balance. Check and tighten the wheel nuts

C0031-76	Left Front Wheel Speed Sensor - Wrong mounting position	- Front left wheel speed sensor incorrectly installed - Incorrect front left wheel speed sensor installed	- Install the front left wheel speed sensor correctly - Install the correct front left wheel speed sensor
C0033-38	Right Front Tone Wheel - Signal frequency incorrect	- Front right wheel speed sensor reluctor ring damaged - Front right wheel bearing failure - Front right wheel/tire size incorrect	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Check the integrity of the front right wheel speed sensor reluctor ring - Check the front right wheel bearing - Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the front right wheel and tire for condition and balance. Check and tighten the wheel nuts
C0034-11	Right Front Wheel Speed Sensor - Circuit short to ground	Front right wheel speed sensor power circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground
C0034-12	Right Front Wheel Speed Sensor - Circuit short to battery	Front right wheel speed sensor power circuit short circuit to power	Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to power
C0034-14	Right Front Wheel Speed Sensor -	Front right wheel speed	Refer to the electrical circuit diagrams and check the front

	Circuit short to ground or open	sensor signal circuit short circuit to ground, open circuit, high resistance	right wheel speed sensor signal circuit for short circuit to ground, open circuit, high resistance. Using the manufacturer approved diagnostic system, check datalogger signal - Right Front Wheel Speed Sensor Input (0x2B07)
C0034-1C	Right Front Wheel Speed Sensor - Circuit voltage out of range	Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0034-25	Right Front Wheel Speed Sensor - Signal shape/waveform failure	- Front right wheel speed sensor signal circuit short circuit to ground - Front right wheel speed sensor failure	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor
C0034-29	Right Front Wheel Speed Sensor - Signal invalid	Front right wheel speed sensor power circuit short circuit to ground, short circuit to power,	

		open circuit, high resistance Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0034-2F	Right Front Wheel Speed Sensor - Signal erratic	Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0034-31	Right Front Wheel Speed Sensor - No signal	Front right wheel speed sensor signal circuit short circuit to	

		ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0034-4A	Right Front Wheel Speed Sensor - Incorrect component installed	Incorrect front right wheel speed sensor installed	Install the correct front right wheel speed sensor
C0034-62	Right Front Wheel Speed Sensor - Signal compare failure	- Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance

C0034-66	Right Front Wheel Speed Sensor - Signal has too many transitions/events	- Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front right wheel/tire size incorrect	- Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the front right wheel and tire for condition and balance. Check and tighten the wheel nuts
C0034-76	Right Front Wheel Speed Sensor - Wrong mounting position	- Front right wheel speed sensor incorrectly installed - Incorrect front right wheel speed sensor installed	- Install the front right wheel speed sensor correctly - Install the correct front right wheel speed sensor
C0036-38	Left Rear Tone Wheel - Signal frequency incorrect	- Rear left wheel speed sensor reluctor ring damaged - Rear left wheel bearing failure - Rear left wheel/tire size incorrect	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Check the integrity of the rear left wheel speed sensor reluctor ring - Check the rear left wheel bearing

			Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the rear left wheel and tire for condition and balance. Check and tighten the wheel nuts
C0037-11	Left Rear Wheel Speed Sensor - Circuit short to ground	Rear left wheel speed sensor power circuit short circuit to ground	Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground
C0037-12	Left Rear Wheel Speed Sensor - Circuit short to battery	Rear left wheel speed sensor power circuit short circuit to power	Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to power
C0037-14	Left Rear Wheel Speed Sensor - Circuit short to ground or open	Rear left wheel speed sensor signal circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, open circuit, high resistance. Using the manufacturer approved diagnostic system, check datalogger signal - Left Rear Wheel Speed Sensor Input (0x2B08)
C0037-1C	Left Rear Wheel Speed Sensor - Circuit voltage out of range	Rear left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0037-25	Left Rear Wheel Speed Sensor - Signal shape/waveform failure	- Rear left wheel speed sensor signal circuit short circuit to ground - Rear left wheel speed sensor	

		failure	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor
C0037-29	Left Rear Wheel Speed Sensor - Signal invalid	- Rear left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0037-2F	Left Rear Wheel Speed Sensor - Signal	Rear left wheel speed sensor	

	erratic	signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0037-31	Left Rear Wheel Speed Sensor - No signal	Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0037-4A	Left Rear Wheel Speed Sensor - Incorrect component installed	Incorrect rear left wheel speed sensor installed	Install the correct rear left wheel speed sensor
C0037-62	Left Rear Wheel Speed Sensor - Signal compare failure	- Rear left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear left wheel	

		speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0037-66	Left Rear Wheel Speed Sensor - Signal has too many transitions/events	- Rear left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear left wheel/tire size incorrect	- Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the rear left wheel and tire for condition and balance. Check and tighten the wheel nuts
C0037-76	Left Rear Wheel Speed Sensor - Wrong mounting	Rear left wheel speed sensor incorrectly	- Install the rear left wheel speed sensor correctly - Install the correct rear left

	position	installed Incorrect rear left wheel speed sensor installed	wheel speed sensor
C0039-38	Right Rear Tone Wheel - Signal frequency incorrect	- Rear right wheel speed sensor reluctor ring damaged - Rear right wheel bearing failure - Rear right wheel/tire size incorrect	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Check the integrity of the rear right wheel speed sensor reluctor ring - Check the rear right wheel bearing - Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the rear right wheel and tire for condition and balance. Check and tighten the wheel nuts
C003A-11	Right Rear Wheel Speed Sensor - Circuit short to ground	Rear right wheel speed sensor power circuit short circuit to ground	Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground
C003A-12	Right Rear Wheel Speed Sensor - Circuit short to battery	Rear right wheel speed sensor power circuit short circuit to power	Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to power
C003A-14	Right Rear Wheel Speed Sensor - Circuit short to ground or open	Rear right wheel speed sensor signal circuit short circuit to ground, open circuit, high	Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, open circuit, high resistance. Using the

		resistance	manufacturer approved diagnostic system, check datalogger signal - Right Rear Wheel Speed Sensor Input (0x2B09)
C003A-1C	Right Rear Wheel Speed Sensor - Circuit voltage out of range	▪ Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C003A-25	Right Rear Wheel Speed Sensor - Signal shape/waveform failure	▪ Rear right wheel speed sensor signal circuit short circuit to ground ▪ Rear right wheel speed sensor failure	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) ▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor
C003A-29	Right Rear Wheel Speed Sensor - Signal invalid	▪ Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear right wheel speed sensor signal circuit short circuit to	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) ▪ Refer to the electrical circuit

		ground, short circuit to power, open circuit, high resistance	diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C003A-2F	Right Rear Wheel Speed Sensor - Signal erratic	Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C003A-31	Right Rear Wheel Speed Sensor - No signal	Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high

			resistance
C003A-4A	Right Rear Wheel Speed Sensor - Incorrect component installed	Incorrect rear right wheel speed sensor installed	Install the correct rear right wheel speed sensor
C003A-62	Right Rear Wheel Speed Sensor - Signal compare failure	- Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C003A-66	Right Rear Wheel Speed Sensor - Signal has too many transitions/events	- Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check that the wheels and

		Rear right wheel/tire size incorrect	tires comply with the manufacturer's specification for the vehicle. Check the rear right wheel and tire for condition and balance. Check and tighten the wheel nuts
C003A-76	Right Rear Wheel Speed Sensor - Wrong mounting position	- Rear right wheel speed sensor incorrectly installed - Incorrect rear right wheel speed sensor installed	- Install the rear right wheel speed sensor correctly - Install the correct rear right wheel speed sensor
C0042-11	Brake Pedal Position Sensor "Circuit A" - Circuit short to ground	Brake pedal position sensor signal 1 circuit short circuit to ground	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 1 circuit for short circuit to ground
C0042-15	Brake Pedal Position Sensor "Circuit A" - Circuit short to battery or open	Brake pedal position sensor signal 1 circuit short circuit to power, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 1 circuit for short circuit to power, open circuit, high resistance
C0042-38	Brake Pedal Position Sensor "Circuit A" - Signal frequency incorrect	Brake pedal position sensor signal 1 circuit short circuit to ground, short	

		circuit to power, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 1 circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0043-11	Brake Pedal Position Sensor "Circuit B" - Circuit short to ground	Brake pedal position sensor signal 2 circuit short circuit to ground	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 2 circuit for short circuit to ground
C0043-15	Brake Pedal Position Sensor "Circuit B" - Circuit short to battery or open	Brake pedal position sensor signal 2 circuit short circuit to power, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 2 circuit for short circuit to power, open circuit, high resistance
C0043-38	Brake Pedal Position Sensor "Circuit B" - Signal frequency incorrect	Brake pedal position sensor signal 2 circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles

			Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 2 circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C0061-24	Lateral Acceleration Sensor - Signal stuck high	Lateral acceleration sensor failure	NOTE: The sensor is located within/integral to the restraints control module Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index
C0061-64	Lateral Acceleration Sensor - Signal plausibility failure	Lateral acceleration sensor failure	NOTE: The sensor is located within/integral to the restraints control module Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index
C0062-64	Longitudinal Acceleration Sensor - Signal plausibility failure	Longitudinal acceleration sensor failure	

			NOTES: ▪ The sensor is located within/integral to the restraints control module ▪ This DTC can be set if the vehicle is being tested on chassis dynamometer rollers ▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index
C0063-64	Yaw Rate Sensor - Signal plausibility failure	▪ Yaw rate sensor failure	NOTES: ▪ The sensor is located within/integral to the restraints control module ▪ This DTC can be set if the vehicle is being tested on chassis dynamometer rollers ▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index
C0064-64	Roll Rate Sensor - Signal plausibility failure	▪ Roll rate sensor failure	

			NOTES: ▪ The sensor is located within/integral to the restraints control module ▪ This DTC can be set if the vehicle is being tested on chassis dynamometer rollers ▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index
C006B-00	Stability System Active Too Long - No sub type information	▪ Wheel bearing failure ▪ Wheel speed sensor reluctor ring fault ▪ High speed CAN bus (chassis or powertrain) circuits short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: If this DTC is logged for the first time, the only action is to clear the DTC. Should the DTC reoccur, follow the actions as described below ▪ Check the integrity of the wheel bearings ▪ Using the manufacturer approved diagnostic system, check datalogger signals - Left Front Wheel Speed Sensor Input (0x2B06) - Right Front Wheel Speed Sensor Input (0x2B07) - Left Rear Wheel Speed Sensor Input (0x2B08) - Right Rear Wheel Speed Sensor Input (0x2B09) ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN

			bus (chassis and powertrain) circuits for short circuit to ground, short circuit to power, open circuit, high resistance
C0072-4B	Brake Temperature Too High - Over temperature	Excessive use of brakes	NOTE: This DTC applies only to hybrid electric vehicles Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest
C100A-19	Brake Pedal Position Sensor - Circuit current above threshold	- Brake pedal position sensor signal 1 circuit short circuit to ground, short circuit to power, open circuit, high resistance - Brake pedal position sensor signal 2 circuit short circuit to ground, short circuit to power, open circuit, high resistance - Brake pedal position sensor signal circuits transposed	NOTE: This DTC applies only to hybrid electric vehicles - Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 1 circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 2 circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the brake pedal position sensor signal circuits for correct connection
C100A-28	Brake Pedal Position Sensor - Signal bias level out of range / zero adjustment failure	- Brake and accelerator pedals applied simultaneously - Brake pedal position sensor not calibrated	NOTE: This DTC applies only to hybrid electric vehicles

			- Using the manufacturer approved diagnostic system, clear the DTCs and retest - Using the manufacturer approved diagnostic system, perform routine - Calibrate Brake Travel Sensor (0x304B)
C100A-54	Brake Pedal Position Sensor - Missing calibration	Brake pedal position sensor not calibrated	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, perform routine - Calibrate Brake Travel Sensor (0x304B)
C100A-64	Brake Pedal Position Sensor - Signal plausibility failure	- Air in brake hydraulic system - Brake hydraulic system leaking - Anti-lock brake system hydraulic control unit failure	NOTE: This DTC applies only to hybrid electric vehicles - Refer to the relevant section of the workshop manual and bleed the brake system - Check the brake hydraulic system for leaks and repair as necessary - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - On Demand Self Test (0x0202). If the fault persists, install a new anti-lock brake system hydraulic control unit
C100D-38	Brake Pedal Position Sensor Circuit 1 - Signal frequency incorrect	Brake pedal position sensor signal 1 circuit short circuit to ground, short circuit to power, open circuit,	NOTE: This DTC applies only to hybrid electric vehicles

		high resistance	Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 1 circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C100E-38	Brake Pedal Position Sensor Circuit 2 - Signal frequency incorrect	Brake pedal position sensor signal 2 circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the brake pedal position sensor signal 2 circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C100F-1C	Brake Pedal Position Sensor Supply - Circuit voltage out of range	Brake pedal position sensor supply circuit short circuit to ground	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the brake pedal position sensor supply circuit for short circuit to ground
C1013-7A	Brake System Pressure - Fluid leak or seal failure	Brake hydraulic system leaking	NOTE: This DTC applies only to hybrid electric vehicles Check the brake hydraulic system for leaks and repair as necessary
C1015-92	Vacuum Supply - Performance or incorrect operation	Electric brake vacuum pump circuit open	

		circuit, high resistance ▪ Electric brake vacuum pump failure ▪ Brake booster vacuum leak	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the electrical circuit diagrams and check the electric brake vacuum pump circuit for open circuit, high resistance ▪ Test the operation of the electric brake vacuum pump ▪ Check the brake booster and pipe for vacuum leaks
C101A-14	Vacuum Pressure Sensor - Circuit short to ground or open	▪ Brake vacuum sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the electrical circuit diagrams and check the brake vacuum sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C101A-1C	Vacuum Pressure Sensor - Circuit voltage out of range	▪ Brake vacuum sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the electrical circuit diagrams and check the brake vacuum sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C101A-64	Vacuum Pressure Sensor - Signal plausibility failure	▪ Brake vacuum sensor circuit short circuit to	

		ground, short circuit to power, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the brake vacuum sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
C101C-12	Brake Vacuum Pump - Circuit short to battery	Electric brake vacuum pump relay circuit short circuit to power	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the electric brake vacuum pump relay circuit for short circuit to power
C101C-14	Brake Vacuum Pump - Circuit short to ground or open	Electric brake vacuum pump relay circuit short circuit to ground, open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles Refer to the electrical circuit diagrams and check the electric brake vacuum pump relay circuit for short circuit to ground, open circuit, high resistance
C101C-17	Brake Vacuum Pump - Circuit voltage above threshold	Electric brake vacuum pump circuit short circuit to power	NOTE: This DTC applies only to hybrid electric vehicles

			Refer to the electrical circuit diagrams and check the electric brake vacuum pump circuit for short circuit to power
C101C-53	Brake Vacuum Pump - Deactivated	- Brake booster vacuum leak - Excessive use of brakes	NOTE: This DTC applies only to hybrid electric vehicles - Check the brake booster and pipe for vacuum leaks - Using the manufacturer approved diagnostic system, clear the DTCs and retest
C101C-92	Brake Vacuum Pump - Performance or incorrect operation	- Brake booster vacuum leak - Electric brake vacuum pump circuit open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles - Check the brake booster and pipe for vacuum leaks - Refer to the electrical circuit diagrams and check the electric brake vacuum pump circuit for open circuit, high resistance
C101F-49	Generic Valve Failure - Internal electronic failure	Anti-lock brake system hydraulic control unit failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system hydraulic control unit
C1029-97	ABS Primary Circuit Pressure lower than Secondary - Component or system operation obstructed or	- Brake system hydraulic leak - Anti-lock brake system hydraulic control unit failure	NOTE: This DTC applies only to hybrid electric vehicles

	blocked		▪ Check the brake system for leaks ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - On Demand Self Test (0x0202). If the fault persists, install a new anti-lock brake system hydraulic control unit
C102E-68	Stability Control System - Event information	▪ Rolling road operation	NOTE: This DTC applies only to hybrid electric vehicles ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest
C1054-68	System Functionality Reduced By Diagnostic Request - Event information	▪ Diagnostic routine DC00 in progress	NOTE: This DTC is set when routine 0xDC00 is performed (Engineering only) ▪ Exit the diagnostic routine
C105C-68	ABS Dynamometer Test Mode Active - Event information	▪ Dynamometer test mode active	NOTES: ▪ This DTC applies only to hybrid electric vehicles ▪ Dynamometer test mode is entered by operating the brake pedal, accelerator pedal and dynamic stability control switch simultaneously for 5 seconds

			Using the manufacturer approved diagnostic system, clear the DTCs and retest
C1109-24	Vehicle Dynamics Control Switch - Signal stuck high	Dynamic stability control switch stuck active	Test the operation of the dynamic stability control switch. If fault persists, install a new terrain response switchpack
C1135-14	Left Rear Brake Pressure Sensor - Circuit short to ground or open	Anti-lock brake system hydraulic control unit failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - On Demand Self Test (0x0202). If the fault persists, install a new anti-lock brake system hydraulic control unit
C1135-28	Left Rear Brake Pressure Sensor - Signal bias level out of range / zero adjustment failure	Anti-lock brake system hydraulic control unit failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - On Demand Self Test (0x0202). If the fault persists, install a new anti-lock brake system hydraulic control unit
C1135-64	Left Rear Brake Pressure Sensor - Signal plausibility failure	- Air in brake hydraulic system - Brake hydraulic system leaking - Anti-lock brake system hydraulic control unit	NOTE: This DTC applies only to hybrid electric vehicles Refer to the relevant section

		failure	of the workshop manual and bleed the brake system - Check the brake hydraulic system for leaks and repair as necessary - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - On Demand Self Test (0x0202). If the fault persists, install a new anti-lock brake system hydraulic control unit
C1A90-12	Wheel Speed Sensor Supply - Circuit short to battery	Wheel speed sensor supply circuit(s) short circuit to power	Refer to the electrical circuit diagrams and check the wheel speed sensor supply circuits for short circuit to power
C1A95-01	Wheel Speed Sensor - General electrical failure	- Wheel speed sensor circuit(s) short circuit to ground, short circuit to power, open circuit, high resistance - Battery/charging system fault	- Refer to the electrical circuit diagrams and check the wheel speed sensor circuits for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the relevant section of the workshop manual and test the battery and charging system
C1A95-62	Wheel Speed Sensor - Signal compare failure	- Wheel speed sensor circuit(s) short circuit to ground, short circuit to power, open circuit, high resistance - Wheel speed sensor reluctor ring fault - Wheel bearing failure - Wheel and/or tire size incorrect	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) - Refer to the electrical circuit diagrams and check the wheel speed sensor circuits for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system,

			check datalogger signals - Left Front Wheel Speed Sensor Input (0x2B06) - Right Front Wheel Speed Sensor Input (0x2B07) - Left Rear Wheel Speed Sensor Input (0x2B08) - Right Rear Wheel Speed Sensor Input (0x2B09) ▪ Check the integrity of the wheel bearings ▪ Check that the wheel and tire sizes are correct to the vehicle specification
C1A95-64	Wheel Speed Sensor - Signal plausibility failure	▪ Wheel speed sensor circuit(s) short circuit to ground, short circuit to power, open circuit, high resistance ▪ Wheel speed sensor reluctor ring fault ▪ Wheel bearing failure ▪ Wheel and/or tire size incorrect	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) ▪ Refer to the electrical circuit diagrams and check the wheel speed sensor circuits for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check datalogger signals - Left Front Wheel Speed Sensor Input (0x2B06) - Right Front Wheel Speed Sensor Input (0x2B07) - Left Rear Wheel Speed Sensor Input (0x2B08) - Right Rear Wheel Speed Sensor Input (0x2B09) ▪ Check the integrity of the wheel bearings ▪ Check that the wheel and tire sizes are correct to the vehicle specification
C1A95-	Wheel Speed Sensor -	▪ Wheel speed	▪ Check the wheel speed

76	Wrong mounting position	sensor(s) incorrectly installed Incorrect wheel speed sensor(s) installed	sensors installation Install a new wheel speed sensor as necessary
C1A99-00	Pressure Sensor - No sub type information	- Air in brake hydraulic system - Brake hydraulic system leaking - Anti-lock brake system hydraulic control unit failure	NOTE: This DTC applies only to hybrid electric vehicles - Refer to the relevant section of the workshop manual and bleed the brake system - Check the brake hydraulic system for leaks and repair as necessary - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - On Demand Self Test (0x0202). If the fault persists, install a new anti-lock brake system hydraulic control unit
C1A99-14	Pressure Sensor - Circuit short to ground or open	Anti-lock brake system hydraulic control unit failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system hydraulic control unit
C1A99-28	Pressure Sensor - Signal bias level out of range / zero adjustment failure	Anti-lock brake system hydraulic control unit failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system hydraulic control unit
C1B00-29	Steering Angle Sensor - Signal invalid	- Steering angle sensor module incorrectly installed - Steering angle	- Check the steering angle sensor module installation - Using the manufacturer approved diagnostic system, check the steering angle

		sensor module fault	sensor module for related DTCs and refer to the relevant DTC index
C1B00-64	Steering Angle Sensor - Signal plausibility failure	- Steering angle sensor module incorrectly installed - Steering angle sensor module failure - Yaw rate sensor incorrect installation	NOTE: This DTC is set when there is conflicting information coming from the steering angle sensor and the yaw rate sensor - Check the steering angle sensor module installation - Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index - Check the yaw rate sensor installation
C1B00-76	Steering Angle Sensor - Wrong mounting position	- Steering angle sensor module incorrectly installed - Steering angle sensor module failure	- Check the steering angle sensor module installation - Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index
C1B00-92	Steering Angle Sensor - Performance or incorrect operation	- Steering angle sensor module incorrectly installed - Steering angle sensor module failure	- Check the steering angle sensor module installation - Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index
C1B02-14	Return Pump - Circuit short to ground or open	Anti-lock brake system hydraulic control unit	

		pump circuit short circuit to ground, open circuit, high resistance ▪ Anti-lock brake system hydraulic control unit failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump circuit for short circuit to ground, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - On Demand Self Test (0x0202). If the fault persists, install a new anti-lock brake system hydraulic control unit
C1B02-16	Return Pump - Circuit voltage below threshold	▪ Anti-lock brake system hydraulic control unit pump circuit short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump circuit for short circuit to ground, open circuit, high resistance
C1B02-49	Return Pump - Internal electronic failure	▪ Anti-lock brake system control module failure	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module
C1B22-	Hill Descent Switch -	▪ Hill descent	▪ Test the operation of the hill

24	Signal stuck high	control switch stuck active	descent control switch. If fault persists, install a new terrain response switchpack
C2009-95	Front Axle Wheel Speed Sensors Swapped - Incorrect assembly	Front wheel speed sensor connectors transposed	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check that the front wheel speed sensors are connected correctly
C200A-95	Rear Axle Wheel Speed Sensors Swapped - Incorrect assembly	Rear wheel speed sensor connectors transposed	NOTE: After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) Refer to the electrical circuit diagrams and check that the rear wheel speed sensors are connected correctly
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive/sequence	Invalid data received from another control	Using the manufacturer approved diagnostic system, check the snapshot data to

	counter incorrect / not updated	module via the high speed CAN bus (powertrain)	determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-83	High Speed CAN Communication Bus - Value of signal protection calculation incorrect	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	Missing message from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	High speed CAN bus (powertrain) circuits short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuits for short circuit to ground, short circuit to power, open circuit, high resistance
U0074-88	Control Module Communication Bus "B" Off - Bus off	High speed CAN bus (chassis) circuits short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuits for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software	The anti-lock brake system	Check the anti-lock brake system control module part

	Incompatibility - No sub type information	control module is not compatible with the vehicle	number, install the correct part as required. Clear the DTCs and retest. To confirm the repair and extinguish the warning lamps, clear the DTC, cycle the ignition state to off, then return the ignition state to on and wait up to 10 seconds
U0405-68	Invalid Data Received From Cruise Control Module - Event information	Missing/invalid data from the adaptive speed control module	Using the manufacturer approved diagnostic system, check the adaptive speed control module for related DTCs and refer to the relevant DTC index
U201A-55	Control Module Main Calibration Data - Not configured	Anti-lock brake system control module is not correctly configured	Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software
U2300-54	Central Configuration - Missing calibration	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. If the fault persists, check and up-date the car configuration file as necessary
U3000-44	Control Module - Data memory failure	Anti-lock brake system control module failure	Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. If the fault persists, install a new anti-lock brake system control module
U3000-46	Control Module - Calibration/parameter memory failure	Anti-lock brake system control module is not correctly configured	Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control

			module
U3000-48	Control Module - Supervision software failure	▪ Anti-lock brake system control module is not correctly configured	▪ Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module
U3000-49	Control Module - Internal electronic failure	▪ Anti-lock brake system control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module
U3000-4B	Control Module - Over temperature	▪ Anti-lock brake system hydraulic control unit valve overheat protection has been activated	NOTE: This DTC may be set if the manufacturer approved diagnostic system has been operating the valves for a prolonged period ▪ Allow sufficient time for the unit to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest
U3000-50	Control Module - Reserved by document	▪ Anti-lock brake system control module failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control

			module
U3000-51	Control Module - Not programmed	Anti-lock brake system control module is not correctly configured	Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module
U3000-52	Control Module - Not activated	Anti-lock brake system control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module
U3000-53	Control Module - Deactivated	Anti-lock brake system control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module
U3000-55	Control Module - Not configured	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary
U3000-	Control Module -	Anti-lock brake	

77	Commanded position not reachable	system hydraulic control unit failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system hydraulic control unit
U3003-17	Battery Voltage - Circuit voltage above threshold	Battery/charging system fault	Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system
U3006-16	Control Module Input Power "A" - Circuit voltage below threshold	- Battery/charging system fault - Anti-lock brake system control module power or ground circuit high resistance	- Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system - Refer to the electrical circuit diagrams and check the anti-lock brake system control module power and ground circuits for high resistance